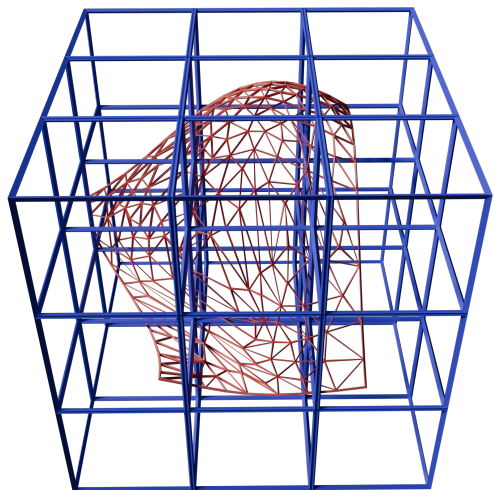


Input

Manifold

① Position



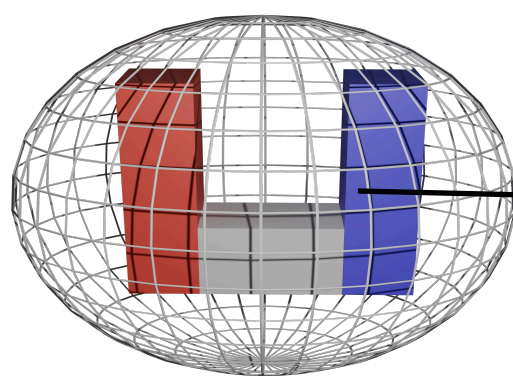
Euclidean

② Position

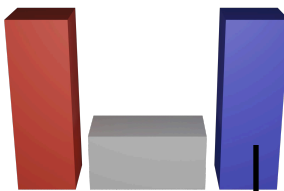
③ Normal

④ View

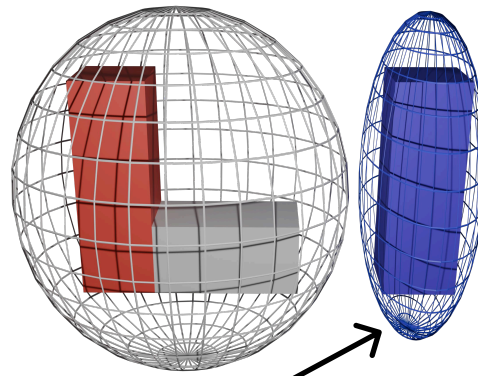
⑤ Adaptive Environment Map



Convex
Decompose



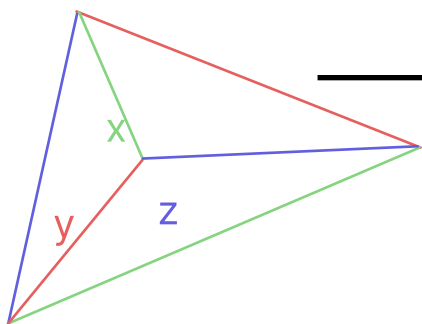
Large Error



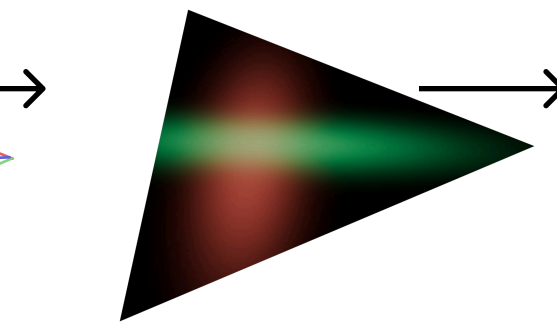
Env Map
Split

On Manifold 2D Gaussian Mixture

①



Barycentric
Coordinates



Probability Density
as Weight

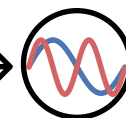
Diffuse

⑥ Color

⑦ Weight

KAN Shader

②



Position
Layer

③

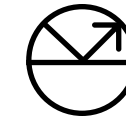
④



Reflection
Layer

③

④



Environment
Layer

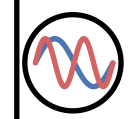
⑤

⑧ Specular
Color

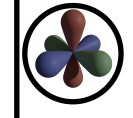
⑥

⑦

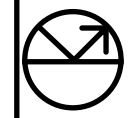
Output



Fourier
Encoding



Spherical
Encoding



Analytical
Env Map
Look Up